

Huawei’s AI Strategy & the HCCDA-AI Certification

Session 2 — Huawei HCCDA-AI Course

Part 1 — From AI Basics to AI-as-a-Service

| Quick Recap: What We Already Know

Last session covered the general AI landscape: AI vs. Machine Learning vs. Deep Learning, the three ways a machine learns (supervised, unsupervised, reinforcement), NLP, Computer Vision, and what a foundation model like ChatGPT actually is.

Today answers the next natural question: **how does a real company actually use any of this, without training their own models?**

| The Global AI Industry: Landscape, Trends & Adoption Barriers

Before narrowing in on Huawei specifically, it helps to see the wider picture — who the major players are, what’s currently driving the industry, and why adoption isn’t as simple as “just add AI.”

The Global Landscape, By Category

Category	Examples
Foundation model developers	OpenAI, Google DeepMind, Anthropic, Meta
Major cloud AI platforms	AWS, Microsoft Azure, Google Cloud, Huawei Cloud, Alibaba Cloud
AI-specialized hardware	NVIDIA (GPUs), Huawei (Ascend chips), AMD
Sector-specific AI applications	Healthcare diagnostics, fintech fraud detection, manufacturing quality control

No single company “owns” AI — the industry is really a stack, with model developers, cloud platforms, hardware makers, and application builders all depending on each other. Huawei plays across several of these layers at once: it builds AI chips (Ascend), a cloud platform (Huawei Cloud), and ready-to-use services (Cloud EI) — which is part of why understanding Huawei’s approach gives you a useful lens on the industry as a whole.

Current AI Trends

- **GenAI (Generative AI)** — models that create new content (text, images, code, audio) rather than just classifying or predicting from existing data. ChatGPT and similar tools are the most visible example.
- **LLMs (Large Language Models)** — the specific type of foundation model behind most GenAI chatbots and writing tools, trained on huge amounts of text.
- **Edge AI** — covered earlier in Part 2 of this course — running models directly on local devices instead of the cloud.
- **Agentic AI** — a newer trend: AI systems that don’t just answer a single question, but take multi-step actions toward a goal (e.g., booking a task across several tools without a human approving every step).
- **Multimodal AI** — models that work across more than one type of data at once (text and images together, for example), rather than being limited to just one.

Adoption Barriers

Even with all this progress, most companies don’t adopt AI as fast or as easily as the hype suggests. Three recurring barriers:

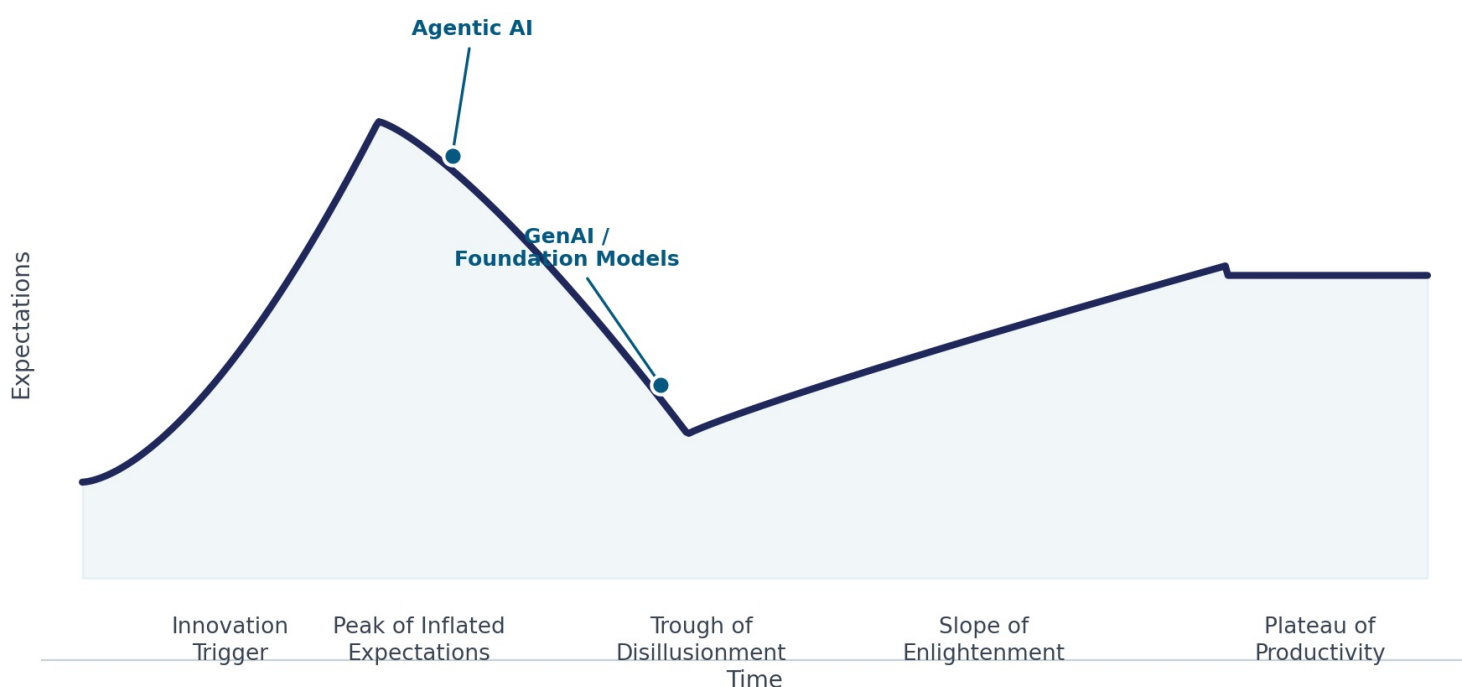
- **Data** — many companies don’t have clean, labeled data ready to use, and collecting it takes time and money
- **Regulation** — data protection and AI-specific laws are still evolving in many countries, creating uncertainty about what’s compliant
- **Cost** — compute, integration, and the specialist talent needed to implement AI responsibly all add up, even when using ready-made cloud AI services

The Hype Cycle: A Tool for Reading AI Trends Realistically

Analyst firm Gartner popularized a framework called the **Hype Cycle** for tracking how new technologies are typically received over time. It has five phases:

1. **Innovation Trigger** — a breakthrough or early proof-of-concept generates interest
2. **Peak of Inflated Expectations** — early excitement leads to overenthusiasm and inflated claims
3. **Trough of Disillusionment** — reality sets in as real-world implementation proves harder than expected, and enthusiasm cools
4. **Slope of Enlightenment** — practical, realistic use cases start to emerge and mature
5. **Plateau of Productivity** — the technology becomes a normal, well-understood part of how things are done

The Hype Cycle Concept — A Simplified Illustration
(with example positions from Gartner's 2026 AI Hype Cycle reporting)



On Gartner's 2026 Hype Cycle, generative AI and foundation models remain in the Trough of Disillusionment — the same phase as the year before — meaning the initial hype has cooled and companies are still working through the real costs and limits of implementation. Agentic AI, meanwhile, became significant enough that Gartner split it out into its own standalone Hype Cycle for the first time in April 2026, placing it at the Peak of Inflated Expectations: only 17% of organizations had actually deployed AI agents at that point, even though more than 60% planned to within two years — the fastest adoption-intent curve Gartner has recorded for any emerging technology. Gartner also predicted that roughly 40% of agentic AI projects would be cancelled by the end of 2027 due to cost overruns, unclear ROI, or governance failures. None of this means these technologies are failing: it's the normal pattern many now-standard tools (cloud computing itself included) went through. It's also a useful mental model for staying grounded when evaluating any new AI claim, including ones you'll hear about Huawei's own products.

Why Companies Don't Build AI From Scratch

Training a model like the one that verifies your face against your CNIC needs three expensive things at once:

- **Huge amounts of labeled data** — often millions of examples, each correctly tagged by a human before training can even start
- **Specialist ML engineers** — people who understand model architectures, training pipelines, and how to debug a model that refuses to learn
- **Serious computing power** — GPU clusters that run for days or weeks per training job, and cost real money every hour they run

A Rough Comparison: Build vs. Buy

	Build Your Own Model	Call a Cloud AI API
Time to first working version	Weeks to months	Minutes to hours
Team needed	ML engineers, data engineers, labelers	A single developer who can write an HTTP request
Upfront cost	High (compute, salaries, data collection)	Low (pay-as-you-go per API call)
Who maintains the model	You	The cloud provider
Flexibility for a very unusual use case	High — full control	Limited to what the API supports

Very few companies — even large ones — have the time, team, and budget for the left column. So instead of building, most companies **rent** the AI capability they need from a cloud provider, the same way they’d rent office space instead of constructing a building.

This is often called **AI-as-a-Service**, and it’s not unique to Huawei — AWS has “Rekognition” and “Comprehend,” Google Cloud has “Vision AI” and “Natural Language API,” and Microsoft Azure has “Cognitive Services.” They all solve the same problem in roughly the same way: send data to an endpoint, get a prediction back. Learning the pattern on Huawei Cloud EI transfers almost directly to any of these.

Worked Example: Estimating the Cost of an Invoice-Reading Feature

Imagine a small logistics company in Lahore wants to automatically read supplier names and totals off scanned invoices, instead of a staff member typing them into a spreadsheet by hand. Here’s a rough, illustrative comparison of what each path might involve:

Building it themselves:

- Collect and label a few thousand scanned invoices (weeks of staff time)
- Hire or contract an ML engineer to build and train a text-recognition model
- Rent GPU compute for training and re-training as new invoice formats appear
- Maintain the model going forward as accuracy drifts

Calling an OCR API instead:

- Sign up for a Huawei Cloud account (same day)
- Write a script that sends each scanned invoice to the OCR API and reads back the extracted text (a day or two of development)
- Pay per invoice processed, with no infrastructure to maintain

For a company whose core business is logistics — not machine learning — the second path gets them a working feature almost immediately, at a fraction of the upfront effort. This is exactly the calculation that pushes most companies toward AI-as-a-Service for common capabilities like OCR, vision, and language processing, and toward building custom models (covered in Part 2) only for the parts of their business that are genuinely unique.

Meet Huawei Cloud EI

EI stands for “Enterprise Intelligence.” It’s Huawei’s catalog of ready-to-use AI services. Instead of training a model, a developer sends a request to Huawei’s already-trained model and gets a result back — this is called **calling an API**.

Think of an API like ordering from a menu instead of cooking the dish yourself: you don’t need to know how the kitchen works, you just need to know what to ask for and how to describe what you want.

For the real mechanics behind that analogy — HTTP methods, endpoints, headers, status codes, and what JSON actually looks like — see the companion **Understanding APIs & JSON** guide, which walks through all of this hands-on in Postman using the Simple Books API.

A Closer Look at Each Service Category

Vision Services: Covers face detection and comparison, object and scene recognition, image tagging, and quality/content moderation (e.g., detecting inappropriate images automatically). A retail app might use this to recognize which product is in a customer’s photo; a security system might use it to flag an unrecognized face at a restricted entrance.

OCR (Optical Character Recognition): Reads printed or handwritten text out of an image and turns it into usable data. Huawei’s OCR services include general text recognition, but also specialized versions — like the “General Table OCR” you’ll use later in this course, tuned specifically for reading rows and columns out of a scanned table (invoices, forms, receipts) rather than

plain paragraphs.

NLP Services: Covers sentiment analysis (is this review positive or negative?), text classification (is this email spam?), entity extraction (pulling names, dates, and amounts out of a block of text), and machine translation.

Speech Services: Covers speech-to-text (transcribing an audio clip), text-to-speech (generating spoken audio from text), and sometimes speaker identification (recognizing *who* is talking, not just *what* they said).

Worked Example: What a Real OCR API Call Looks Like

Huawei's **General Table OCR** service — the one this course focuses on — works like this. A developer sends a `POST` request to an address (an “endpoint”) that looks roughly like:

```
POST https://{endpoint}/v2/{project_id}/ocr/general-table
```

The request includes an authentication token in the header, and either a base64-encoded image or an image URL in the body:

```
Headers:
X-Auth-Token: <your access token>
Content-Type: application/json

Body:
{
  "url": "https://your-storage-bucket/invoice-scan.png"
}
```

Huawei's servers process the image and send back a structured JSON response describing what was found — separating plain text areas from table areas, and giving a confidence score for each detected block of text:

```
{
  "words_region_count": 2,
  "words_region_list": [
    {
      "type": "text",
      "words_block_count": 1,
      "words_block_list": [
        { "words": "INVOICE #4521", "confidence": 0.998 }
      ]
    },
    {
      "type": "table",
      "words_block_count": 2,
      "words_block_list": [
        { "words": "Total", "confidence": 0.991 },
        { "words": "PKR 12,000", "confidence": 0.987 }
      ]
    }
  ]
}
```

Notice the shape: `words_region_list` separates the *type* of content (plain text vs. table), and each block carries both the recognized text and a confidence score, so your code can decide whether to trust a low-confidence read or flag it for a human to check. This nested structure — labeled fields grouped together, lists of similar items — is the same basic pattern you'll meet again when we get to Python: real API responses are just a structured, textual version of that same idea.

Mapping EI Services to What We Already Learned

Huawei EI Service	Which Concept It Is	Local Example
Vision (face / object detection)	Computer Vision	Unlocking your phone by looking at it
OCR (Optical Character Recognition)	Computer Vision	A bank scanning your CNIC to auto-fill your details
NLP (text understanding, sentiment)	Natural Language Processing	A bank or telecom chatbot reading your question
Speech-to-text	NLP	YouTube auto-generating captions live
Text-to-speech	NLP	A GPS app reading directions out loud
Machine translation	NLP	A shopping app auto-translating a product description

The key insight: you already understood these concepts last session. Huawei EI is simply *the specific, callable product version* of ideas you already know.

| Case Study: A Small Retailer’s AI Journey

To see how all of this fits together, follow a fictional example: **Bazaar Mart**, a small Pakistani retail chain with a handful of stores.

The problem: Bazaar Mart’s head office spends hours every week manually entering supplier invoice totals into their accounting system, and store managers have no easy way to tell when shelves are running low without physically walking the aisles.

Step one — the easy win. Bazaar Mart starts with the OCR worked example above: supplier invoices are scanned and sent to Huawei’s General Table OCR API, which extracts the totals automatically. No model training, no ML team — just a script calling an API, live within days.

Step two — expanding to vision. Encouraged by the quick win, Bazaar Mart adds a Vision API call to their mobile stock-check app: a store employee photographs a shelf, and the API tags which products are visible, flagging shelves where expected items are missing.

We’ll pick this same example back up in Part 2, once HiLens and ModelArts are on the table — because Bazaar Mart’s next problem (recognizing their own private-label packaging, which no general-purpose model has ever seen) needs more than an off-the-shelf API.

Part 2 — ModelArts, HiLens, the Certification, and the Job Market

| Beyond APIs: ModelArts, In Detail

Sometimes an off-the-shelf EI service isn’t a perfect fit — you need to recognize a specific product line, a specific handwriting style, or a specific defect type that Huawei’s general-purpose model was never trained on. That’s what **ModelArts** is for: a cloud workbench for building a custom model without starting from zero.

The typical ModelArts workflow looks like this:

1. **Collect data** — gather example images, text, or records relevant to your specific problem
2. **Label the data** — tag each example with the correct answer (manually, or with ModelArts’s auto-labeling assist)
3. **Choose a training approach** — either **AutoML/ExeML** (Huawei automatically picks and tunes a model for you — faster, less control) or a **custom training job** (you write the training code yourself — slower, full control)
4. **Train** — ModelArts runs the training job on Huawei’s compute infrastructure
5. **Evaluate** — check accuracy, precision, recall, and look at examples the model got wrong
6. **Deploy** — publish the trained model as a callable API endpoint, just like the ready-made EI services

This means ModelArts and Cloud EI aren’t really separate worlds — a model you train in ModelArts ends up being called the exact same way (an API request) as a ready-made EI service. The difference is just *who trained the model*.

Worked Example: Applying the Workflow to Bazaar Mart

Continuing the case study — Bazaar Mart’s private-label packaging (their own brand of rice, tea, and spices) doesn’t look like

anything a general-purpose Vision API was trained on. Here's how the six-step ModelArts workflow would apply:

1. **Collect data** — store staff photograph each private-label product from several angles and lighting conditions, building up a few hundred example images per product
2. **Label the data** — each photo is tagged with the correct product name
3. **Choose a training approach** — since the team has no in-house ML engineer, they choose AutoML/ExeML rather than writing custom training code
4. **Train** — ModelArts trains a custom image-classification model on Bazaar Mart's labeled photos
5. **Evaluate** — the team checks accuracy on a held-out batch of photos and notices the model struggles to tell two similar tea packages apart, so they add more labeled examples of those two products specifically
6. **Deploy** — the finished model is published as an API endpoint, and the same mobile stock-check app now recognizes Bazaar Mart's own products, not just generic categories

| Beyond APIs: HiLens, In Detail

HiLens solves a different problem: what happens when you need an instant answer and can't afford the delay of sending data to the cloud and waiting for a response — or when there's no reliable internet connection at all?

HiLens deploys AI models directly onto **edge devices** — typically cameras or small computing units placed on-site. Common architecture:

- A camera captures a video feed continuously
- The AI model (already downloaded onto the device) processes each frame **locally**
- Only the *result* (e.g., "person detected," "empty shelf detected") is sent onward — not the raw video

Real-world HiLens-style use cases:

- A retail store detecting when a shelf is empty and needs restocking
- A factory floor camera detecting a worker not wearing a hard hat
- A parking lot camera counting available spaces in real time
- A smart city camera detecting traffic congestion patterns

Worked Example: Bazaar Mart's Shelf-Monitoring Camera

Back to the case study one more time. Bazaar Mart's mobile stock-check app (from Part 1) works, but it still relies on a staff member remembering to walk the aisles and take photos. Their next step is a HiLens camera mounted above each aisle:

- The camera runs Bazaar Mart's custom shelf-recognition model (built in ModelArts) directly on the HiLens device
- Every few minutes, it checks whether expected products are visible on the shelf
- When a shelf appears empty, it sends a short alert — just the message "Aisle 3, Shelf 2: restock needed" — to the store manager's phone, not a live video stream

This is the tradeoff HiLens is built for: the device doesn't have the raw computing power of the cloud, so the model has to be smaller and more specialized — but the result is instant, and it works even if the store's internet connection drops.

| Why This Matters Locally

Huawei's cloud-AI footprint has grown fastest across Asia, Africa, the Middle East, and Latin America — the same category of market Pakistan's digital economy sits in. The pattern you're learning shows up across:

- **Banks & fintechs** (JazzCash, Easypaisa) — fraud detection, ID verification
- **NADRA-style ID systems** — face and document matching
- **Telecoms** (Jazz, Zong, Ufone) — support chatbots, sentiment analysis on customer feedback
- **Retail & e-commerce** (Daraz) — visual search, recommendations, exactly the pattern in the Bazaar Mart case study above
- **Manufacturing and textiles** — a large part of Pakistan's export economy — quality inspection using vision models on the production line
- **Healthcare** — assisting (not replacing) radiologists by flagging X-rays or scans that may need a closer look
- **Education** — automatically grading multiple-choice sheets with OCR, or translating learning material into regional languages

A transferable skill: the pattern you'll practice — sign up, get an access key, call an API, read the response — is nearly identical across Huawei Cloud, AWS, Azure, and Google Cloud. Learning it once on Huawei Cloud EI makes the others much easier to pick up later, which matters if you end up job-hunting on platforms that list any of these cloud skills. Entry-level postings in this space often list titles like "AI Solutions Associate," "Cloud Support Engineer," or "Junior ML/AI Developer" — roles where a hands-on, vendor-recognized credential like HCCDA-AI is a meaningful signal even before years of experience.

| What Is the HCCDA-AI Certification, Really?

HCCDA-AI = Huawei Certified Cloud Developer - Artificial Intelligence. It’s an entry-level, vendor-specific credential. Passing it demonstrates you can:

- Explain core AI/ML concepts
- Carry out a basic requirement analysis for an AI feature
- Call and use Huawei Cloud EI API services
- Describe what ModelArts and HiLens are for and when to use each

To an employer, it’s evidence you can actually work with cloud-AI tools hands-on — not just discuss AI in the abstract.

How the Certification Process Works

1. **Online learning** — the course content itself
2. **Lab exercises** — hands-on practice, which can be retried an unlimited number of times before the real exam
3. **Theory + lab exam** — a combined exam with one attempt, testing both concepts and practical skill
4. **Certificate** — valid for 3 years once earned

Question counts and passing scores are set by Huawei directly and are confirmed on their certification page closer to your exam date, since these details are occasionally revised between exam versions.

The 6 Exam Domains, and Where We’ll Cover Each

#	Domain	Where We’ll Cover It
1	Intro to AI & Huawei’s AI Strategy	Sessions 1-2 (today)
2	AI Application Requirement Analysis	Upcoming session
3	Huawei Cloud EI – API Services	Upcoming sessions
4	Introduction to Huawei Cloud ModelArts	Upcoming sessions
5	Huawei Cloud HiLens Service	Upcoming sessions
6	AI Application Product Testing & Optimization	Upcoming session

| Quick Quiz

1. What does “EI” stand for in Huawei Cloud EI?
2. Name the five phases of the Gartner Hype Cycle, in order.
3. Name one Huawei EI service that matches the “Computer Vision” concept from last session.
4. Why would a company choose to call a cloud AI API instead of training its own model?
5. In the OCR worked example, what two things does each recognized block of text carry alongside the text itself?
6. In the Bazaar Mart case study, why did they need ModelArts instead of just the ready-made Vision API?
7. What’s the core difference between ModelArts and HiLens?
8. Is the HCCDA-AI exam a single attempt or unlimited attempts? What about the lab exercises?

| Before Next Session

- Create a Huawei Cloud account (a personal account is enough to start)
- Re-read the glossary from last session
- Come ready to make your first real API call next session

| Glossary Additions

Term	Meaning in plain words
API	A way for your code to “call” a service someone else built and get a result back
Huawei Cloud EI	Huawei’s collection of ready-to-use AI services (vision, NLP, speech, OCR)
ModelArts	Huawei’s platform for training and deploying a custom AI model
HiLens	Huawei’s service for running AI models on edge devices (cameras, etc.) instead of only in the cloud
Edge AI	Running AI directly on a local device rather than sending data to the cloud and waiting for a response
Endpoint	The web address an API request is sent to
Confidence score	A number showing how sure a model is about a particular prediction
Hype Cycle	Gartner’s five-phase framework for how new technologies are typically received over time, from early hype to realistic, everyday use
GenAI / LLM	Generative AI (models that create new content) and Large Language Models (the text-based models behind most GenAI chatbots)

Answer Key

1. **Enterprise Intelligence**
2. **Innovation Trigger → Peak of Inflated Expectations → Trough of Disillusionment → Slope of Enlightenment → Plateau of Productivity**
3. **Vision** or **OCR**
4. It avoids the cost, time, and specialist expertise needed to build and train a model from scratch — the company can use an already-trained model instantly instead.
5. A **type** (text or table) and a **confidence score**, alongside the recognized text itself.
6. Because their private-label packaging is unique to their business — no general-purpose model was ever trained to recognize it, so they needed to train a custom model on their own labeled photos.
7. **ModelArts** trains or fine-tunes custom models in the cloud. **HiLens** deploys models onto edge devices (like cameras) for fast, local, real-time inference.
8. The **exam** is one attempt. The **lab exercises** can be retried an unlimited number of times before the exam.